

Current to Voltage Converter & Comparator

ENGS152 Circuits Lab

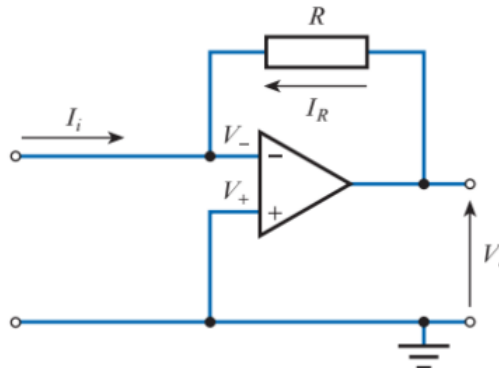
2 May 2026

Step 1: Current to voltage converter

We will use an OpAmp to measure the current and convert it to a voltage. The output is given by the formula

$$V_o = I_i R$$

The topology is shown below



Below you can see the calculation of the R value we got by using the formula mentioned above.

Photoresistor – from 0 to 1mA

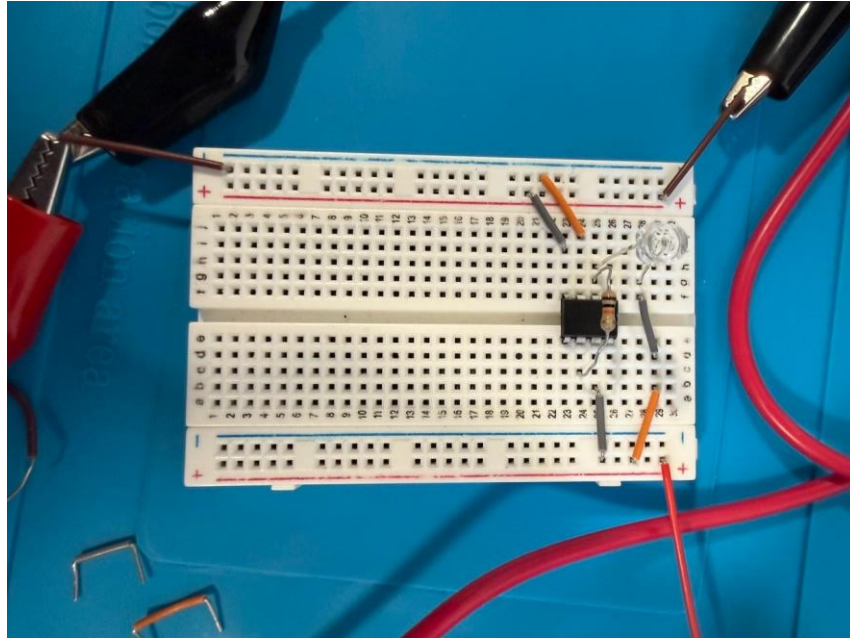
Output voltage – from 0 to -9V

$$V_o = -I_i R$$

$$R = \frac{V_{max}}{I_{i max}} = \frac{9}{0.001} = 9k\Omega$$

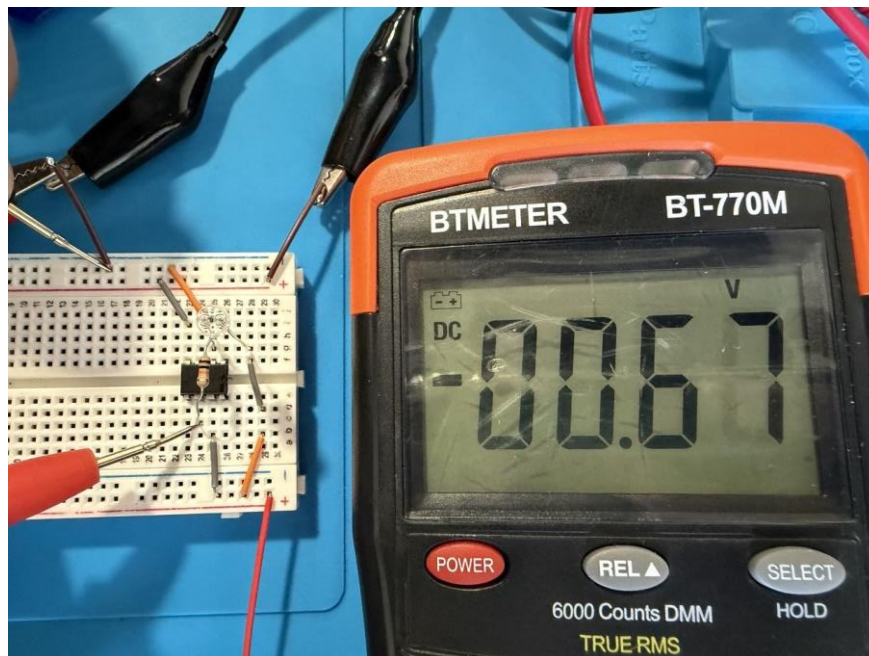
We will choose $10\text{k}\Omega$ as in the lab we have that value available, but luckily the measured value was close to ours, it was $9.6\text{k}\Omega$.

Below you can see our real life circuit:

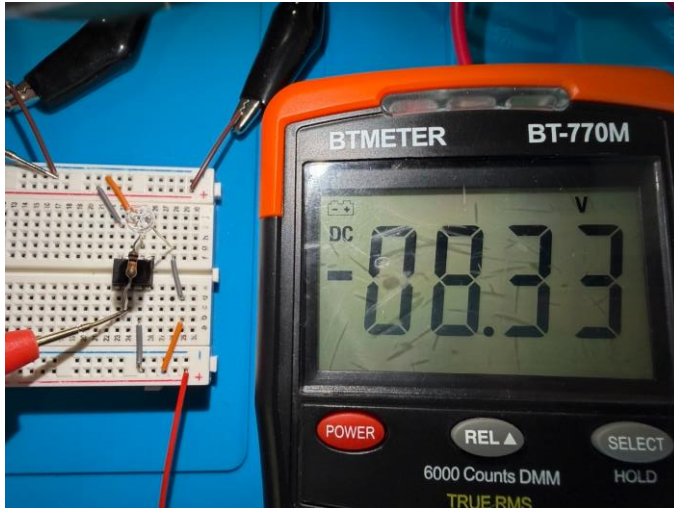


Now you can see that when we don't have any light on the phototransistor we have almost 0V in the output. When we have the light on the phototransistor, we see that it reaches nearly -8.3V

Low Luminosity



High Luminosity



Step 2: Comparator

In this step, we will use the OpAmp as a comparator. It will compare the voltages on the Inverting and Non-inverting terminals.

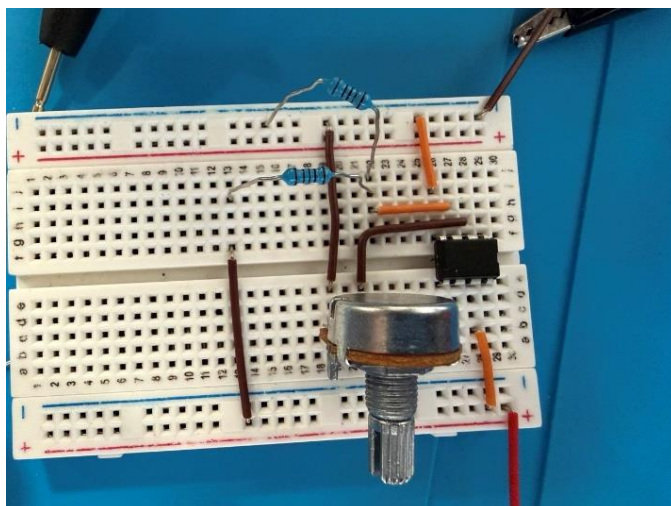
When $V_{inv} < V_{non_inv}$, Output = V_{+}

When $V_{inv} > V_{non_inv}$, Output = 0V

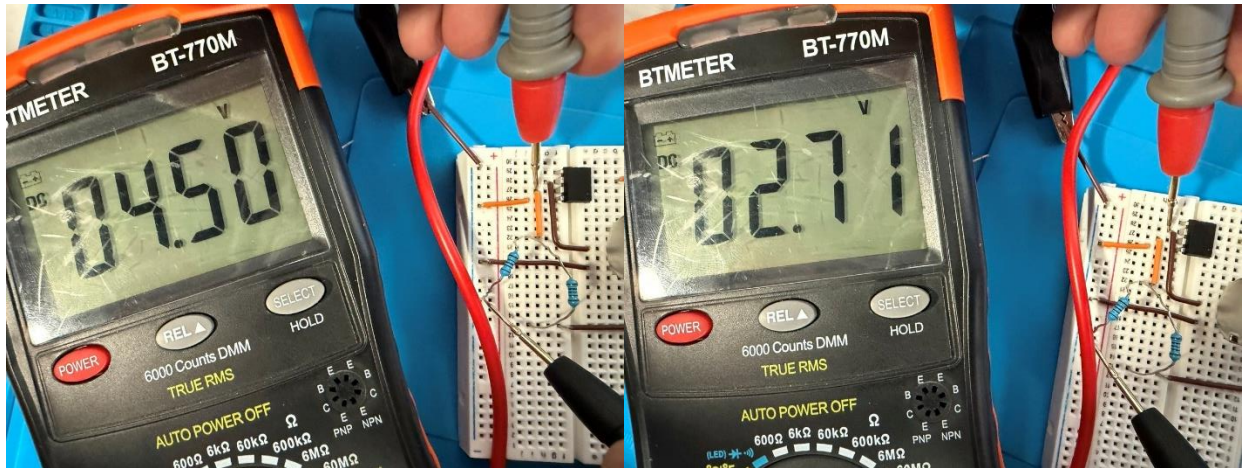
We will build two voltage dividers. For that we will use 2 fixed resistors and one potentiometer (0-10 k Ω).

The R values here we choose 1k Ω each, as they are equal, the voltage will be halved.

Now you can see our circuit and each case in the pictures.

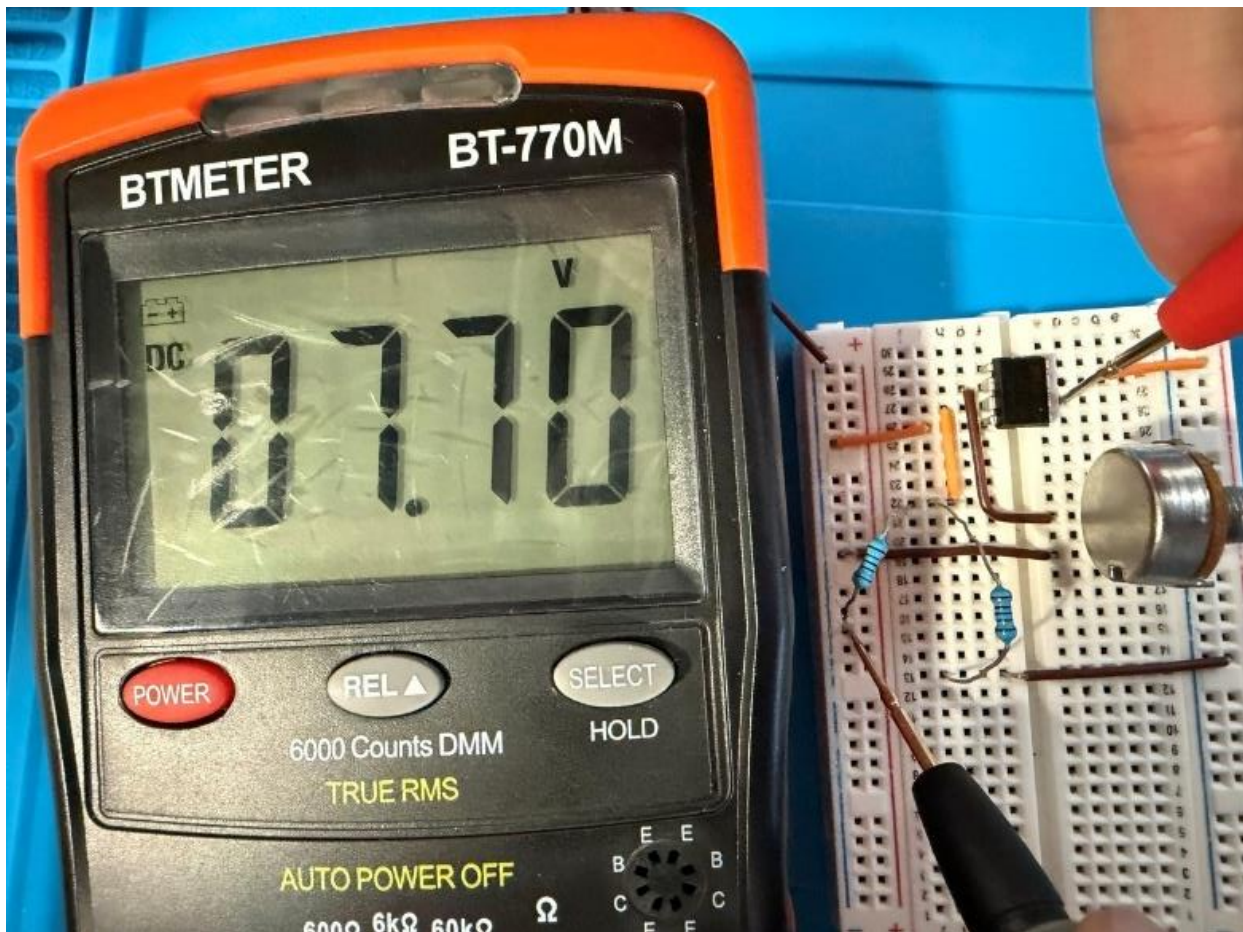


Case 1: $V_{INV} < V_{NON_INV}$



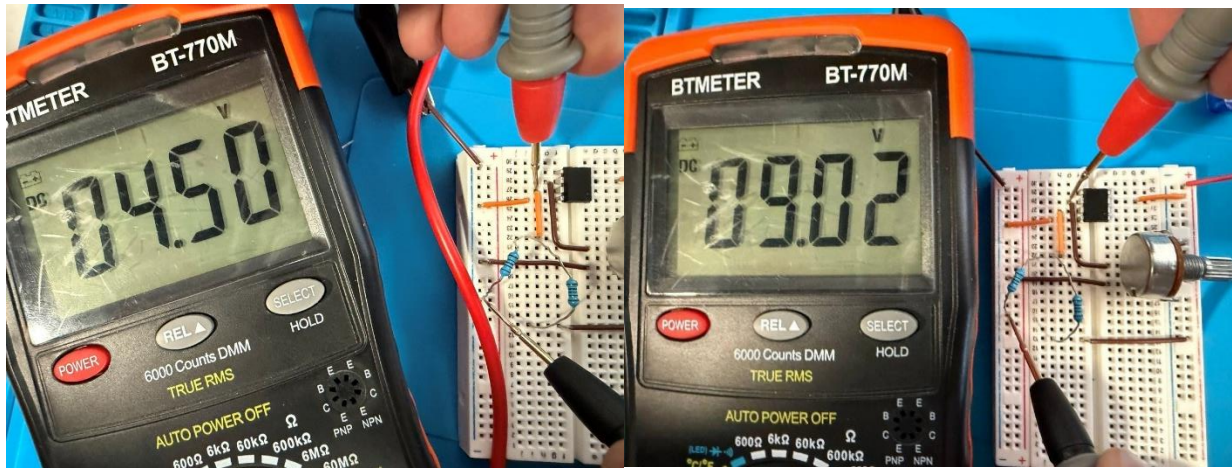
$V_{NON_INV} = 4.5 \text{ V}$

$V_{INV} = 2.71 \text{ V}$



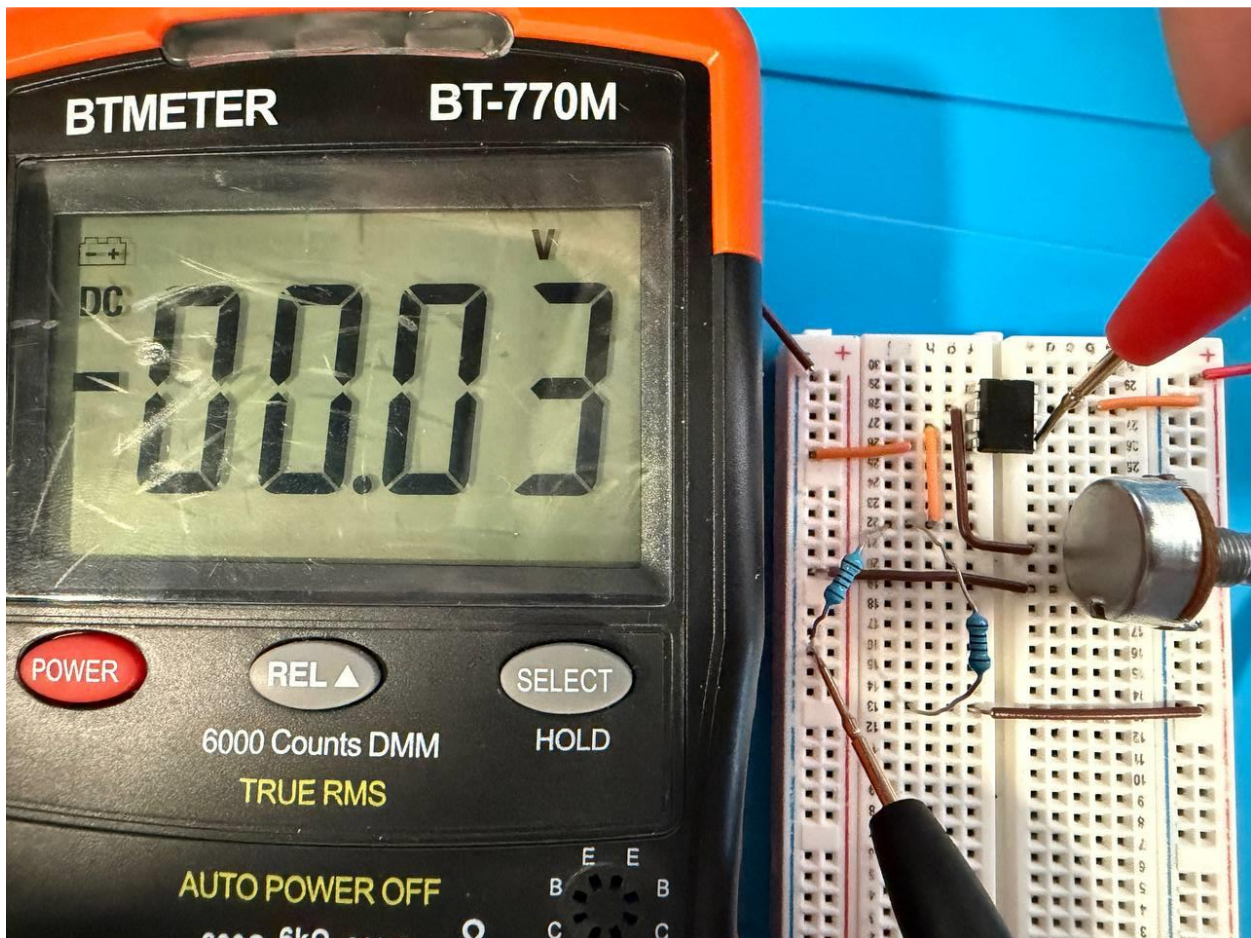
$V_{OUTPUT} = 7.70 \text{ (Positive)}$

Case 2: $V_{INV} > V_{NON_INV}$



$$V_{NON_INV} = 4.5 \text{ V}$$

$$V_{INV} = 9.02 \text{ V}$$



$$V_{OUTPUT} = 0.03 \approx 0 \text{ V (GND)}$$